

Experiment 1: Prompt Adherence

Comprehensive Analysis Report

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Research Question: To what extent do LLM message roles (System, User, Assistant) affect the LLM's prompt adherence?

1. Simple Schema Results

Dataset: Simple JSON extraction tasks | Samples: 200 | Model: openai/gpt-4.1-mini

Variant	Valid/Total	Success %	Extraneous Text %
user_only	200/200	100.0%	0.00%
system_plus_user	200/200	100.0%	0.00%
user_plus_assistant_seed	198/200	99.0%	0.00%

2. Complex Schema Results

Dataset: Complex schemas (nested objects, arrays, conditional, union) | Samples: 100 | Model: openai/gpt-4.1-mini

Variant	Valid/Total	Success %	Extraneous Text %
user_only	100/100	100.0%	0.00%
system_plus_user	100/100	100.0%	0.00%
user_plus_assistant_seed	90/100	90.0%	2.63%

Schema Type Breakdown

Schema Type	Overall	user_only	system_plus_user	user_plus_assistant_seed
Nested Arrays	248/258 (96.1%)	86/86 (100%)	86/86 (100%)	76/86 (88.4%)
Nested Objects	42/42 (100%)	14/14 (100%)	14/14 (100%)	14/14 (100%)

3. Comparative Analysis

Variant	Simple	Complex	Difference
user_only	100.0%	100.0%	+0.0%
system_plus_user	100.0%	100.0%	+0.0%
user_plus_assistant_seed	99.0%	90.0%	-9.0%

4. Key Findings

Finding 1: Message Role Structure Has Minimal Impact

Both user_only and system_plus_user achieved identical 100% success rates on both simple and complex schemas. System messages do not provide any advantage for JSON schema adherence.

Finding 2: Schema Complexity Does Not Reduce Adherence

Both user_only and system_plus_user maintained 100% success even with complex schemas (nested objects, arrays). Modern LLMs handle complex JSON structures exceptionally well.

Finding 3: Assistant Seeding Trade-off

user_plus_assistant_seed showed a 9% drop in success rate with complex schemas (99% → 90%). Assistant seeding is effective but has reliability trade-offs with complexity.

Finding 4: Schema Type Differences

Nested objects: 100% success across all variants. Nested arrays: 96.1% success (mainly due to assistant seeding issues). Arrays are slightly more challenging, but not significantly.

5. Conclusions

- 1. Message role structure (System vs User) has MINIMAL impact on JSON schema adherence. Both user_only and system_plus_user achieved 100% success on both simple and complex schemas.
- 2. Assistant seeding is effective but less reliable with complex schemas, showing a 9% drop (99% → 90%).
- 3. Schema complexity does NOT significantly reduce adherence for standard prompts. Both variants maintained 100% success rate even with nested structures.
- 4. For JSON schema adherence tasks, simple user-only prompts are sufficient. System messages do not provide additional benefit.
- 5. Role structure differences may be more apparent in adversarial scenarios (Experiment 2) or long-context decision-making (Experiment 3).

6. Statistical Summary

Total Samples Tested: 600 (200 simple + 300 complex)
Overall Success Rate: 98.8%
- Simple schemas: 99.7% (598/600)
- Complex schemas: 96.7% (290/300)

Variant	Simple	Complex	Overall
user_only	200/200 (100%)	100/100 (100%)	300/300 (100%)
system_plus_user	200/200 (100%)	100/100 (100%)	300/300 (100%)
user_plus_assistant_seed	198/200 (99%)	90/100 (90%)	288/300 (96%)